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MATH 455: Partial Differential Equations

Assignment #2 - More Fourier Series
Due Thursday, September 17, 2026

1. (12 points) Define the function

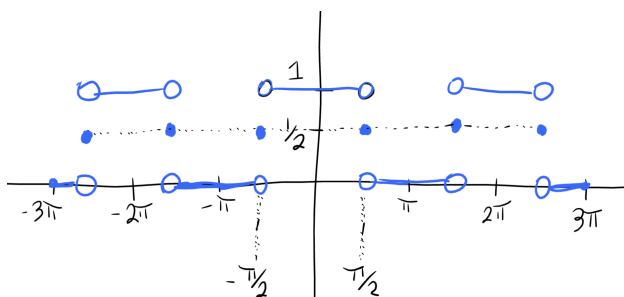
$$f(x) = \begin{cases} 1, & \text{if } |x| < \frac{\pi}{2} \\ 0, & \text{if } |x| > \frac{\pi}{2}, \end{cases}$$

and let $F(x)$ be the Fourier series of $f(x)$ on the interval $(-\pi, \pi)$. Sketch by hand $F(x)$ on the interval $(-3\pi, 3\pi)$ and compute a formula for $F(x)$ using the Fourier coefficients.

Solution — Since f is not periodic on $\mathbb{R}$, we restrict its domain to $[-\pi, \pi]$ and fill in left-right-averaged discontinuities:

$$f_{2\pi}(x) = \begin{cases} 1 & \text{if } |x| < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < |x| \leq \pi \\ \frac{1}{2} & \text{if } |x| = \frac{\pi}{2} \end{cases}$$

Now let $g(x)$ be the periodic extension of $f_{2\pi}(x)$ to all $\mathbb{R}$. By the Fourier convergence theorem, $F(x) = g(x)$ for all $\mathbb{R}$, so sketching $F(x)$ on $(-3\pi, 3\pi)$ is equivalent to sketching $g(x)$ on $(-3\pi, 3\pi)$:



We compute the Fourier series of $g(x)$:

$$\begin{aligned}
a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) dx = 1 \\
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \sin(nx) dx = 0 \quad (\text{by oddness of product } g(x) \sin(nx)) \\
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \cos(nx) dx \\
&= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(nx) dx \\
&= \frac{1}{\pi} \left[\frac{1}{n} \sin(nx) \right]_{-\pi/2}^{\pi/2} \\
&= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right)
\end{aligned}$$

Hence

$$\begin{aligned}
F(x) &= \frac{1}{2} + \sum_{n=0}^{\infty} \left[\frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right] \cos(nx) \\
&= \frac{1}{2} + \frac{2}{\pi} \sum_{n=0}^{\infty} \left[\frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(nx)
\end{aligned}$$

Since

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & \text{if } n \text{ even} \\ 1 & \text{if } n \equiv 1 \pmod{4} \\ -1 & \text{if } n \equiv 3 \pmod{4} \end{cases}$$

we want to skip the even terms, hopping over the odd terms instead, and alternating between a 1 and -1 coefficient, starting with 1:

$$F(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos((2n+1)x)$$

2. (8 points) We know that if $f(x)$ is odd on the interval $(-\pi, \pi)$, then its Fourier series is composed of only sine functions. What additional symmetry condition on f will make the sine coefficients with even indices become zero (i.e., $b_{2n} = 0$ for all $n \in \mathbb{Z}$)? Provide a specific example of a function with this property and show that it satisfies the condition.

Solution — Let $f(x)$ be odd. We want to find additional conditions on $f(x)$ that will force $b_n = 0$ for even n , or equivalently, to force $b_{2n} = 0$ for all n :

$$\begin{aligned} b_{2n} &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(2nx) dx = 0 \\ &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin(2nx) dx \quad (\text{by evenness of } f(x) \sin(2nx)) \end{aligned}$$

Note that $\sin(2nx)$ is odd about $x = \frac{\pi}{2}$:

$$\begin{aligned} &\sin\left(2n\left(\frac{\pi}{2} + h\right)\right) \\ &= \sin(n\pi + 2nh) \\ &= \sin(n\pi) \cos(2nh) + \cos(n\pi) \sin(2nh) \\ &= -(\sin(n\pi) \cos(2nh) - \cos(n\pi) \sin(2nh)) \\ &= -\sin(n\pi - 2nh) \\ &= -\sin\left(2n\left(\frac{\pi}{2} - h\right)\right) \end{aligned}$$

$x = \frac{\pi}{2}$ is the midpoint of our integration range $[0, \pi]$, so the $[0, \pi]$ -integral of any function that is odd about $x = \frac{\pi}{2}$ must be zero. Hence, for $\int_0^{\pi} f(x) \sin(2nx) dx$ to be zero, it is sufficient for $f(x)$ to be **even about** $x = \frac{\pi}{2}$ (in addition to being odd about $x = 0$).

One such function that is odd (about $x = 0$) and even about $x = \frac{\pi}{2}$ is

$$\boxed{f(x) = \sin(x)}$$

$$\begin{aligned} \sin(h) &= \sin(0 + h) \\ &= \sin(0) \cos(h) + \cos(0) \sin(h) \\ &= -\sin(0) \sin(h) + \cos(0) \sin(h) \\ &= -(\sin(0) \sin(h) - \cos(0) \sin(h)) \\ &= -\sin(0 - h) \\ &= -\sin(-h) \end{aligned}$$

$$\begin{aligned} \sin\left(\frac{\pi}{2} + h\right) &= \sin\left(\frac{\pi}{2}\right) \cos(h) + \cos\left(\frac{\pi}{2}\right) \sin(h) \\ &= \sin\left(\frac{\pi}{2}\right) \cos(h) - \cos\left(\frac{\pi}{2}\right) \sin(h) \\ &= \sin\left(\frac{\pi}{2} - h\right) \end{aligned}$$

3. (12 points) Let

$$f(x) = x \cos(x), \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$g(x) = x \cos(x), \quad x \in (-1, 1),$$

and

$$h(x) = \begin{cases} 0, & \text{if } 1 < x < 3 \\ 1, & \text{if } -1 < x < 1 \\ x, & \text{if } -3 < x < -1 \end{cases}$$

be periodic functions (defined via periodic extension) for $x \in \mathbb{R}$. For each function

- (a) Determine whether or not the function is in PC^1 on the interval over which it is defined.

Solution — $f(x) = x \cos(x) \in PC^1[-\pi/2, \pi/2]$ because both x and $\cos(x)$ are in $C^\infty(\mathbb{R})$, so their product must be in $C^\infty(\mathbb{R})$, which implies $f(x) \in PC^1[-\pi/2, \pi/2]$.

$g(x) = x \cos(x) \in PC^1[-1, 1]$ for an identical reason.

$h(x) \in PC^1[-3, 3]$ because it is built from finitely-many slices of 0, 1, and x , all three of which are in $C^\infty(\mathbb{R})$ (which is sufficient to imply that their first derivatives approach finite values at all the junction points).

- (b) State the function to which its Fourier series (defined on the interval: $[-\frac{\pi}{2}, \frac{\pi}{2}]$ for $f(x)$, $[-1, 1]$ for $g(x)$, and $[-3, 3]$ for $h(x)$) converges

Note: Do not compute the Fourier series (or coefficients) to do this!

Solution — We use the Fourier convergence theorem, which asserts that for a function $f \in PC^1[a, b]$, its Fourier series shall converge to the midpoint $(f(a^+) + f(a^-))/2$ of the left and right limits at each point $x = a$.

Since $f(x) \in PC^1[-\pi/2, \pi/2]$ and $f(-\pi/2) = f(\pi/2)$ its Fourier series $F(x)$ is exactly the periodic extension of $f(x)$ to all $x \in \mathbb{R}$:

$$F(x) = x \cos(x) \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \text{With } F(x + n\pi) := F(x) \text{ for } n \in \mathbb{Z}$$

Since $g(x) \in PC^1[-1, 1]$ and $g(-1) \neq g(1)$, its Fourier series $G(x)$ is the periodic extension of $g(x)$ to all $x \in \mathbb{R}$ but with left-right-averaged jump discontinuities filled in (left endpoint $-\cos(1)$ and right endpoint $\cos(1)$ averaging to 0):

$$G(x) = \begin{cases} x \cos(x), & \text{if } |x| < 1 \\ 0, & \text{if } |x| = 1 \end{cases} \quad \text{With } G(x + 2n) := G(x) \text{ for } n \in \mathbb{Z}$$

Similarly, since $h(x) \in PC^1[-3, 3]$ the Fourier series of $h(x)$ converges to the periodic extension of $h(x)$ with left-right-averaged jump discontinuities filled in:

$$H(x) = \begin{cases} 0, & \text{if } 1 < x < 3 \\ \frac{1}{2}, & \text{if } x = 1 \\ 1, & \text{if } -1 < x < 1 \\ 0, & \text{if } x = -1 \\ x, & \text{if } -3 < x < -1 \\ -\frac{3}{2}, & \text{if } |x| = 3 \end{cases} \quad \text{With } H(x + 6n) := H(x) \text{ for } n \in \mathbb{Z}$$

(c) Sketch by hand three periods of the Fourier series in part (b).

Solution — A

4. (8 points) Let

$$F(x) = \sum_{n=1}^{\infty} \frac{(-1)^n n^{3/2} + 2}{n^8 + 2n^6 + 1} \cos(nx)$$

How many continuous derivatives can we be sure that F possesses? Justify your answer using tools from Calculus II.

Solution — A